

Number Guessing Game (C++)

CloudExify C++ Internship – Project 1

A simple console-based Number Guessing Game developed in C++. The player selects a difficulty level, guesses a randomly generated number, and the program keeps track of the best score using file handling.

Developer Information

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Features

- Three difficulty levels:
 - Easy (1–500)
 - Medium (1–100)
 - Hard (1–200)
- Random number generation using `rand()` and `srand()`
- Input validation
- Best score saved using file handling (`bestscore.txt`)
- Replay option after each game
- Displays the number of attempts
- Modular code using separate functions

Technologies Used

- C++
- File Handling (`fstream`)
- Standard Library
- Console Application

Project Structure

Number-Guessing-Game/

|— main.cpp

|— bestscore.txt

|— README.md

└─ screenshots/

 └─ gameplay1.png

 └─ gameplay2.png

How to Run

##Using g++

1. Clone the repository

```
```bash
```

```
git clone https://github.com/yourusername/Number-Guessing-Game.git
```

```
```
```

2. Open the project folder

```
```bash
```

```
cd Number-Guessing-Game
```

```
```
```

3. Compile the program

```
```bash
```

```
g++ main.cpp -o game
```

```
```
```

4. Run the executable

****Windows****

```bash

game.exe

```

****Linux/macOS****

```bash

./game

```

##Program Workflow

1. Start the program.
2. Select a difficulty level.
3. Enter your guesses.
4. The program provides hints whether the secret number is higher or lower.
5. Guess the correct number.
6. Your best score is automatically saved.
7. Choose whether to play again.

Screenshots

Main Menu

```
===== NUMBER GUESSING GAME =====  
Best Score :7 Attempts  
  
Select Difficulty Level  
1. Easy    (1-50)  
2. Medium  (1-100)  
3. Hard    (1-200)  
==-- Default range is 100 --==
```

Winning the Game

```
Enter Guess: 22  
    Number is lower! Guess high  
Enter Guess: 27  
    Number is lower! Guess high  
Enter Guess: 30  
    Number is lower! Guess high  
Enter Guess: 32  
    Number is higher! Guess low  
Enter Guess: 31  
  
Congratulations You guessed correctly  
Attempts: 6  
  
*** New Best Score Saved! ***
```

Concepts Used

- Functions
- Loops
- Conditional Statements
- Switch Case
- Random Number Generation

- File Handling
- User Input Validation

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